

Programmable Eclairs

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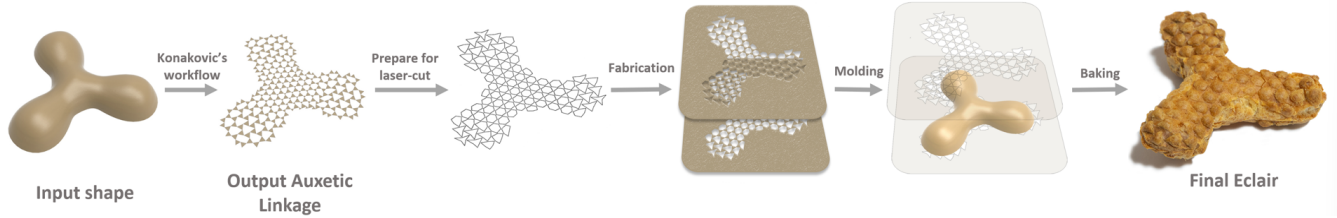


Figure 1: The workflow of baking a programmable éclair.

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1 INTRODUCTION

Computational food fabrication has recently become a topic of interest, using various fabrication approaches for different edible materials (see e.g., [Yang et al. 2021] and references within). In this work, we propose to leverage a recent approach for generating deployable doubly curved surfaces using auxetic patterns [Konaković-Luković et al. 2018] to fabricate eclairs of different shapes. In previous work the doubly curved shape was achieved by using gravity or inflation by air. We, on the other hand, show how to use the expansion of dough during baking as the driving force that yields the shape.

Specifically, we use the auxetics-based workflow [Konaković-Luković et al. 2018] to generate a 2D cutting scheme for laser cutting. This scheme is applied to the bottom part of a standard paper baking mold, which is then used as a cover for the baked dough (see Figure 1 and the attached video). During baking, the dough rises differently at different regions of the shape, since it encounters varying resistance depending on the pattern of the laser-cut baking mold. We experimented with a variety of configurations, and demonstrate some preliminary results.

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2 METHOD

Our workflow is described in Figure 1:

- (1) Given the input shape we compute a 2D auxetic pattern which deploys into the input shape [Konaković-Luković et al. 2018]. The computation is done in Grasshopper for Rhino [McNeel and Associates 2022] (Section 2.1).
- (2) We thicken the lines of the 2D pattern to make the computed auxetic linkage valid for laser-cutting.
- (3) We laser-cut a standard paper baking mold according to the pattern we computed.
- (4) We place dough and the baking mold according to the chosen baking method (Section 2.2).
- (5) Finally, We bake the éclair. During baking the dough expands and pushes the 2D auxetic pattern of the laser-cut baking mold into the open state by rotating the triangles of the pattern. Different regions of the shape expand at different rates, thus shaping the éclair.

2.1 Computational Workflow

To compute the 2D auxetic linkage, we use Grasshopper for Rhino. The required input shape is given as a triangle mesh and the output is a 2D auxetic linkage. The computational scheme is based on the workflow of Konaković-Luković et al. [2018]. The Grasshopper code receives as input the triangle mesh and its conformal mapping to the plane (which we compute using BFF [Sawhney and Crane 2017]). Next, we compute a regular triangular meshing on top of the conformal flattening, lift the regular meshing back into 3D using the inverse map and compute the initial linkage in 3D. We optimize the 3D linkage using the Kangaroo2 package [Piker 2021] to get the auxetic pattern in its fully 3D open state. To get the 2D auxetic linkage, we apply the linkage connectivity to the 2D regular meshing to get a linkage on the plane. For the final result, we optimize the edge lengths of the planar linkage to be identical to



Figure 2: Baking setups: without boundary (upper left); dough boundary (upper right); baking weights (beans) boundary (bottom left); and a two-sided baking mold (bottom right).

the edge lengths of the 3D linkage (using kangaroo2). The output from the Grasshopper code is a 2D auxetic linkage that opens into the input shape when activated by the expanding dough.

2.2 Fabrication experiments

We experimented with different baking approaches for fixing the boundary of the baked shape, choosing the amount of dough, etc.

2.2.1 Baking Setups. We tried four setups in our experiments:

Basic setup. The laser-cut paper baking mold is on top of the dough without fixing the boundary. See Figure 2 (upper left).

Dough boundary. The boundary of the input shape is traced with dough. See Figure 2 (upper right).

Weights boundary. Beans are used as weights to fix the boundary. See Figure 2 (bottom left).

Two-sided mold. The dough is covered from top and bottom with paper baking molds which are stapled together, and baking cups are used for support. In this setup the eclair obtains a shape which is a back-to-back gluing of the input shape and its mirror reflection through the base plane. Gravity assists the dough in the bottom part to expand further. See Figure 2 (bottom right).



Figure 3: Different amount of dough. From left to right: smallest to biggest amount of dough. The top (bottom) row shows the eclairs before (after) extracting them from the mold, respectively.

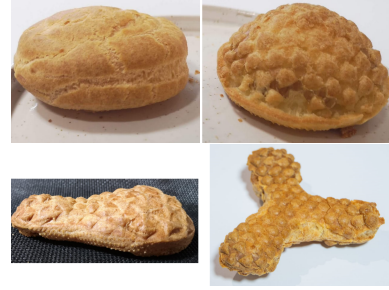


Figure 4: Top: A half sphere shape, without (left) and with (right) our computed mold. Bottom: eclairs examples baked with our approach.

2.2.2 Dough Amount. Extruding too much dough might tear through the triangles of the auxetic linkage mold. On the other hand, not enough dough may lead to insufficient dough rising, and thus the triangles of the auxetic linkage do not reach their maximal opening state and do not obtain the required input shape (see Figure 3).

3 PRELIMINARY RESULTS AND FUTURE WORK

Figure 4 shows some preliminary results for basic input shapes. We have demonstrated a proof-of-concept for using auxetic patterns for shaping baked dough. So far, our computation has been purely geometric, without taking into consideration the physical process of the rising dough. We believe that by modeling more accurately the amount of dough material, and considering the physics of the problem in addition to the geometry, we will be able to achieve more consistent results, and more complicated shapes. Specifically, the amount of dough extruded, as well as the pattern of the starting position of the dough should also be computed.

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